Postural Responses to Mediolateral Rotations: Contributions of Surface, Vision, and Cognitive Load

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***Abstract*— Maintaining postural stability is a complex process influenced by sensory, motor, and cognitive factors. Dynamic balance tests provide valuable insight into these mechanisms by challenging balance under controlled conditions. This study investigated the effects of surface type, visual availability, and cognitive load on postural control during mediolateral rotational disturbances in healthy young adults. Fifteen participants completed trials on a robotic balance platform under eight conditions combining rigid or foam surfaces, eyes open or closed, and with or without a cognitive task. During each trial, ML platform rotations were applied while Center of pressure (COP) responses were recorded. COP data were analyzed across position, velocity, and frequency domains to capture both static and dynamic aspects of postural control. Analysis of variance revealed that vision significantly affected all COP domains, surface type primarily influenced dynamic and frequency measures, and cognitive load altered static and frequency measures. No significant interactions were observed, indicating that each factor contributed independently to balance regulation. These findings suggest that surface, vision, and cognitive load each impose distinct challenges to postural control during ML disturbances, highlighting the importance of evaluating multiple task constraints when assessing balance performance.**

Keywords—Postural Control, Dynamic postural assessment,Mechanical Disturbance, Center of Pressure, COP Measures

I. Introduction

a Postural control is a fundamental aspect of human movement, and its assessment is critical for understanding balance deficits and preventing falls [1]. Maintaining upright stance is a complex task that requires the integration of sensory inputs [2], motor coordination [3], and cognitive resources [4]. While traditional clinical tests are simple and reliable for assessing physical performance, they lack the evaluation of neurophysiological causes of postural instability.

One common method for investigating balance is static posturography, in which participants stand quietly on a force platform while Center of Pressure (COP) displacements are recorded and analyzed. These measures have been extensively studied and provide valuable insights into postural control under unperturbed conditions [5], [6], [7], [8]. However, quiet standing represents only a limited aspect of postural control, as daily activities often involve continuous adjustments and responses to unexpected disturbances. For this reason, dynamic balance assessment, defined as postural control under nonstatic conditions, has become increasingly important [9].

Several dynamic testing methods have been introduced, including shoulder pushes [10], sagittal-axis balance boards [11], functional reach tests [12], [13], and sudden platform rotations [14]. Among these, perturbation-based approaches are particularly powerful because they directly challenge the postural control system. By introducing controlled disturbances, researchers can probe the mechanisms of balance and identify impairments not evident during static tasks. Perturbation paradigms have successfully been applied to distinguish individuals with Parkinson's disease from healthy controls [14], and more broadly to differentiate between populations with neurological or age-related balance disorders such as back pain [9], [13].

In the present study, we focused on mediolateral (ML) disturbances. Participants were exposed to ML platform rotations, and balance was quantified using COP measures that captured static, dynamic, and frequency-domain characteristics of postural control. To further approximate real-world challenges, we systematically manipulated surface type, visual availability, and cognitive load during perturbation trials. This approach provides a comprehensive framework for understanding how multiple factors shape balance responses to mechanical disturbances.

II. Materials and Methods

A. Participants and COP data

Fifteen healthy adult participants (9 females, 6 males; mean age = 24.3 ± 2.6 years; mean height = 166.6 ± 6 cm; mean weight = 64 ± 6 kg) voluntarily took part in this study. All participants reported no history of neuromuscular disorders or balance impairments.

Participants stood barefoot on the platform in eight experimental conditions defined by a combination of three within-subject factors: surface type (rigid vs. foam), visual condition (eyes open vs. closed), and cognitive load (present vs. absent). The cognitive load was induced by instructing the participant to count aloud from a random number in randomly determined step sizes (e.g., counting up from 121 in steps of 7). In each trial, a mechanical perturbation lasting 1 second was applied, followed by a 5-second post-perturbation period during which Center of Pressure (COP) data were recorded. Trials with incorrect behavior (e.g., stepping or holding the support bar) were excluded and repeated.

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A custom-designed robotic balance platform was used to apply controlled mechanical perturbations in the mediolateral (M/L) direction and to simultaneously record COP data (**Fig. 1**). The balance platform included four load cells embedded in a Wii Balance Board configuration, whose signals were amplified using HX711 modules and acquired through an Arduino Due microcontroller .



Fig. 1. Left: A test subject who is interacting with the robot while stading on the foam surface with eyes closed; Right: The robotic balance platform and its element.

The robotic platform was equipped with two AC servo motors, allowing sinusoidal perturbations to be applied by oscillating the platform laterally. A dedicated program implemented in the Arduino environment generated specific perturbation signals, while real-time COP data were collected and transmitted via serial communication.

COP signals were recorded in the form of time series COPx, COPy, time, corresponding respectively to the mediolateral (ML), anterior-posterior (AP), and sampling time. All COP measurements were obtained at an average sampling rate of approximately 86.5 Hz, and a dedicated Python interface facilitated the synchronization of data recording with the perturbation protocol.

Table I. Definition of the COP measures grouped in position, velocity and frequency domains.

Domain	Sway measure
Position	$\text{MD} = \frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert$
	$\text{MAXD} = \max\lvert x(t)\rvert$
	$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
Velocity	$\text{MV} = \frac{1}{N-1}\sum_{i=1}^{N-1}\lvert \dot{x}_i\rvert$
	$\text{RMSV} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}\lvert \dot{x}_i\rvert^2}$
	$\text{MFREQ} = \frac{\text{MV}}{4\sqrt{2}\text{MD}}$
Frequency	$\mu_0 = \sum_{i=1}^{N}\text{PSD}(f_i)$
	$f_{50} = \inf\{i^* \in \mathbb{N}, \sum_{i}^{i^*}\text{PSD}(f_i) \geq 0.5\mu_0\}$
	$f_{95} = \inf\{i^* \in \mathbb{N}, \sum_{i}^{i^*}\text{PSD}(f_i) \geq 0.95\mu_0\}$

B. Data processing and COP Measures

The acquired COP data of valid trials of the subjects was filtered using a 3rd-order Butterworth low-pass filter with a cutoff frequency of 10 Hz to remove the high-frequency noise. The mean of each signal was subsequently subtracted from the signal. Nine COP measures [15], as given in Table I, were calculated from the COP trajectories.

The measures defined were all calculated in ML AP and radial directions. Excluding the radial direction of RMSV, a total of 26 numbers were extracted. Each measure was calculated per trial and averaged across three trials per condition for every participant.

C. Statistical analysis

To assess the impact of surface compliance (rigid/foam), vision (eyes open/closed), and cognitive load (presence/absence) on COP measures, a three-way repeated-measures ANOVA was conducted for each extracted measure. Three significance levels were defined at $\alpha = 0.05$.

III. Results

The main effects of each of the three experimental factors, as well as their interaction effects, were evaluated using p-values obtained from the three-way repeated measures ANOVA. The detailed results of this analysis are presented in Table II`.

The three-way repeated measures ANOVA showed that surface type, vision condition, and cognitive load each had statistically significant main effects on COP measures. Vision significantly affected all three groups of the measures: position, velocity, and frequency. Surface had significant effects in dynamic and frequency measures, while cognitive load influenced static and frequency domains. These results confirm that each experimental factor independently impacted postural control in domain-specific ways without indicating directionality.

All two-way interactions (surface × vision, surface × cognitive load, vision × cognitive load) and the three-way interaction failed to meet the defined significance levels. Therefore, no interaction effects between the experimental factors were statistically significant. Although a few near-significant trends were observed, none were retained for

Table II. ANOVA test results, * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ (S: surface, V: vision, CL: cognitive load, R: radius)

Domain	Measure	Direction	S	V	CL	S×V	S×CL	V×CL	S×V×CL
Position	MD	ML		***	**				
		AP		***					
		R		***	**				
	MaxD	ML		**	*				
		AP		***					
		R		***					
	RMS	ML		***	**				
		AP		***					
		R		***	**				*
Velocity	MV	ML	***	***					
		AP		***					*
		R	***	***					
	RMSV	ML	***	***					
		AP	***	***					
Frequency	MFREQ	ML	*	***	**				
		AP	**	*					
		R	***	*	**				
	μ0	ML		***	**				
		AP		***					
		R		***	**				
	f50	ML	*	**	***		***		
		AP		*					
		R							
	f95	ML	***	***	*		*		
		AP	*	**					
		R	**	**	**				

interpretation. These results suggest that each factor has an independent impact on the COP outcomes.

IV. Discussion

A. Analysis of Results

The results of this study are consistent with Zhang et al. [16], who examined the combined effects of vision and cognitive load on center of pressure (COP) displacement during a ball-hitting perturbation task. In that study, fourteen healthy young adults stood on a custom-built force platform while holding a pot into which a sandbag was suddenly dropped, inducing anterior–posterior perturbations. Vision (eyes open vs. closed) significantly altered COP variability and maximum displacement. Importantly, no interaction between vision and cognitive load was observed.

In contrast to the findings of the present study, Zhang et al. reported that cognitive load, manipulated through subtraction tasks, did not significantly affect COP outcomes in either the anticipatory or compensatory phases. The results of the underlying control strategies engaged during balance recovery.

Overall, the results indicate that in healthy young adults, vision, surface compliance, and cognitive load independently affect postural control during mediolateral disturbance rejection, with effects confined to specific COP measure domains. The findings highlight the importance of considering task type and perturbation direction when comparing across balance studies and suggest that methodological context is critical for interpreting sensory–motor–cognitive interactions in postural control research.

current study also contrast with several quiet stance studies [17], [18], [19] that reported compounded sensory–cognitive effects. A major methodological distinction is that the latter works examined unperturbed quiet stance, whereas the present study evaluated mechanical disturbance rejection. Rapid perturbations engage long-latency reflexes and cortical processes that differ from those used in quiet stance, which may alter how sensory and cognitive inputs are integrated during balance control [20].

Additionally, the present study focused on mediolateral perturbations. According to Peyré-Tartaruga et al. [21], Nordic walking shows that increased mediolateral stability demands are met by enhanced trunk coordination, supporting the idea that in directions with reduced margins of stability, hip–trunk control becomes increasingly critical. However, Zhang et al. [16] applied anterior–posterior perturbations, which rely more on ankle strategies. Thus, the discrepancy between their findings and the present results may be explained, at least in part, by directional differences in the

B. Limitations

This study has several limitations that should be considered when interpreting the findings. The participants consisted of only 15 healthy young adults, which limits the generalizability of the results to other age groups or populations with balance impairments. Additionally, the experimental setup focused exclusively on mediolateral perturbations, leaving the effects of anterior–posterior or multidirectional challenges unexplored. The study also relied solely on COP-based measures, without incorporating complementary biomechanical or neuromuscular data such as

joint kinematics or muscle activation, which could provide deeper insights into underlying control strategies.

Future work should repeat this three-factor analysis during quiet stance to directly compare perturbed and unperturbed conditions, and under mechanical disturbances in anterior–posterior and compound directions. Such extensions could reveal whether independence persists or interactions emerge under different control strategies, contributing to a more complete understanding of sensory–motor–cognitive integration in postural control.

V. Conclusion

This study demonstrated that surface type, vision condition, and cognitive load each exert independent, domain-specific effects on postural control during mediolateral mechanical perturbations in healthy young adults. Vision influenced all COP domains, surface affected velocity and frequency domains, and cognitive load impacted position and frequency domains. No interaction effects were observed, indicating additive rather than cross-factor influences under the tested conditions.

These results highlight the importance of task type and perturbation direction when interpreting balance studies, as findings from quiet stance or anterior–posterior perturbations may not generalize to mediolateral disturbance rejection.

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